

Lab Report 7 - Cybersecurity Analytics

System Logging Protocol (Syslog)

Introduction

This lab is designed to work with VM machines to get practical skills in system logging. Initially I found that both Windows and Kali VMs have same IPv4 addresses and then I set up our VMs to have new IPv4 addresses. Syslog plays a key role in today's network and system administration, and it provides reliable mechanisms for recording, auditing, and troubleshooting system activities. Regardless of what operating system you are using, the operating system has logs, and those logs can be critical in any forensic investigation (Easttom, 2023).

Implementing rsyslog in Kali Virtual machine demonstrates how log events such as SSH authentication attempts can be captured and filter them to use for forensic investigations. Logging is a reactive tool; it allows you to view the record of an event after it has taken place (Andress, 2019). We can maintain accountability and track both failed and successful login attempts by combining syslog with firewalls (ufw) and secure remote access via sshd. The objective of this lab is to install and configure rsyslog, create SSH filters, and verify logging captures to get idea about how the logging events can be used by the security analysts to find security incidents.

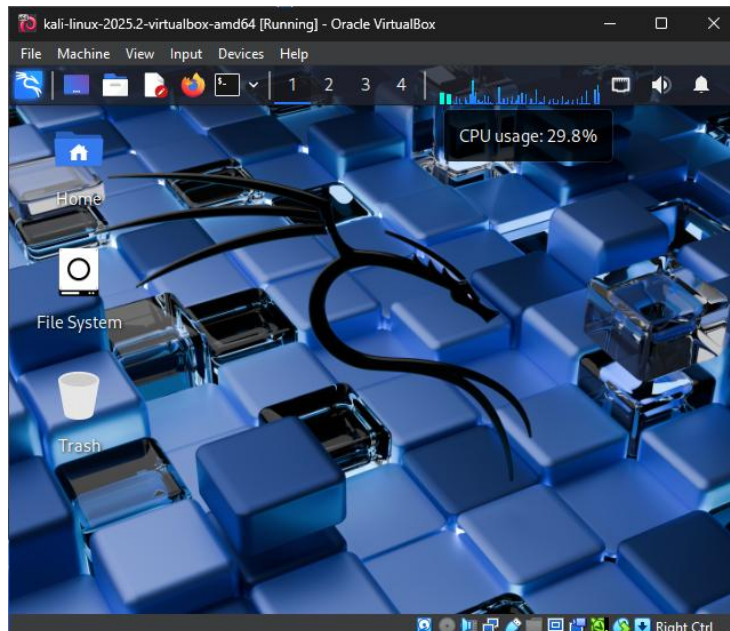
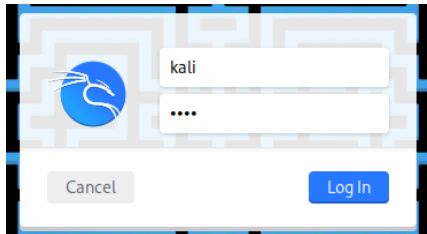
Step 1 - Providing different IPv4 Addresses to the Windows VM and Kali VM

1. Logged to Azure Lab environment using following web link and logged into the Kali VM using Oracle VirtualBox.

https://labs.azure.com/virtualmachines?feature_vnext=true

Username - kali

Password - kali

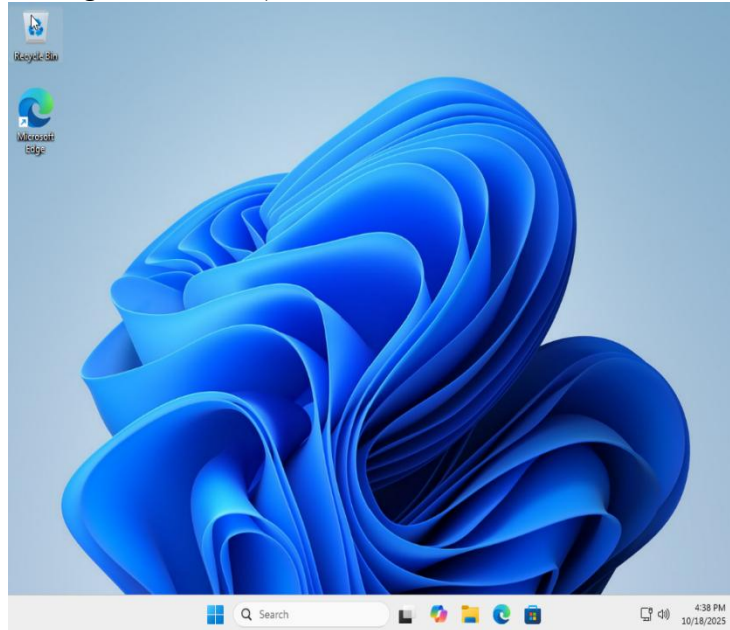


2. Logged to Azure Lab environment using following web link and logged into the Windows VM using Oracle VirtualBox.

https://labs.azure.com/virtualmachines?feature_vnext=true

Username - vboxuser

Password - password (Asked to change the old PW)



3. Checked the IPv4 addresses on Windows VM and the Kali VM. Used **ipconfig** command on Windows VM, and **ifconfig** command in Kali, use

```
C:\Users\vboxuser>ipconfig

Windows IP Configuration

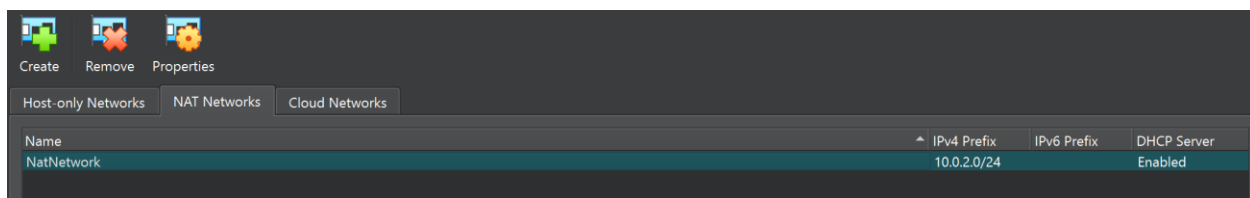
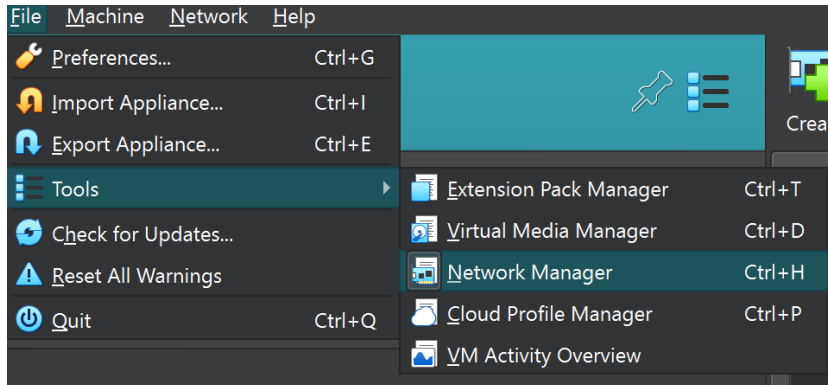
Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : v4kuvoelbxsulmmw0zbp1sinuc.gx.internal.cloudapp.net
    IPv6 Address. . . . . : fd00::b3f3:79d5:20a2:702
    Temporary IPv6 Address. . . . . : fd00::446f:2566:5056:a68
    Link-Local IPv6 Address . . . . . : fe80::ac8f:aff7:66ed:dd50%5
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%5
                               10.0.2.2
```

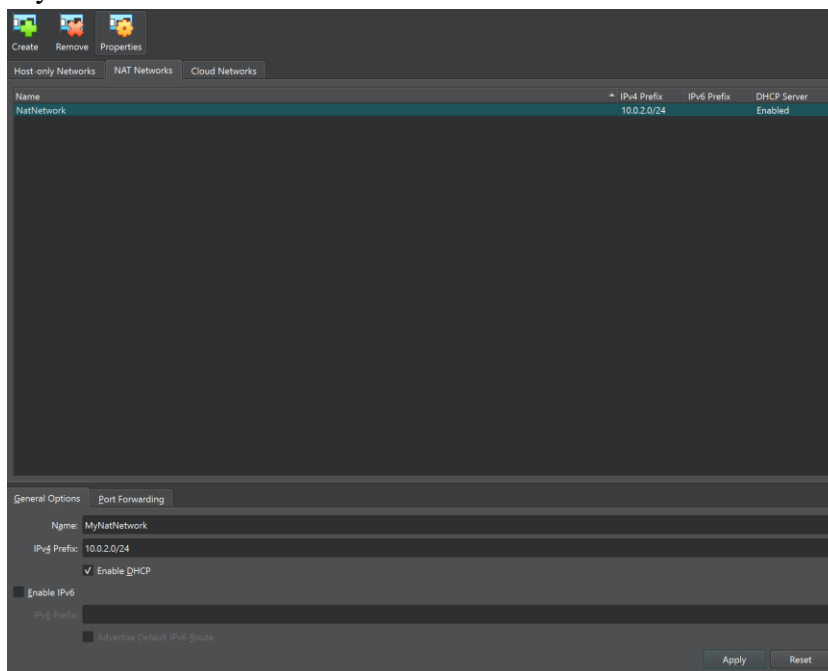
```
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::c1cf:9e72:107:566 prefixlen 64 scopeid 0x20<link>
    inet6 fd00::263e:2c2b:2b3c:9eeb prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:d1:f8:5d txqueuelen 1000 (Ethernet)
    RX packets 9 bytes 4330 (4.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 30 bytes 5531 (5.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Step 2 - Changing the IP Addresses of the Windows VM and the Kali VM

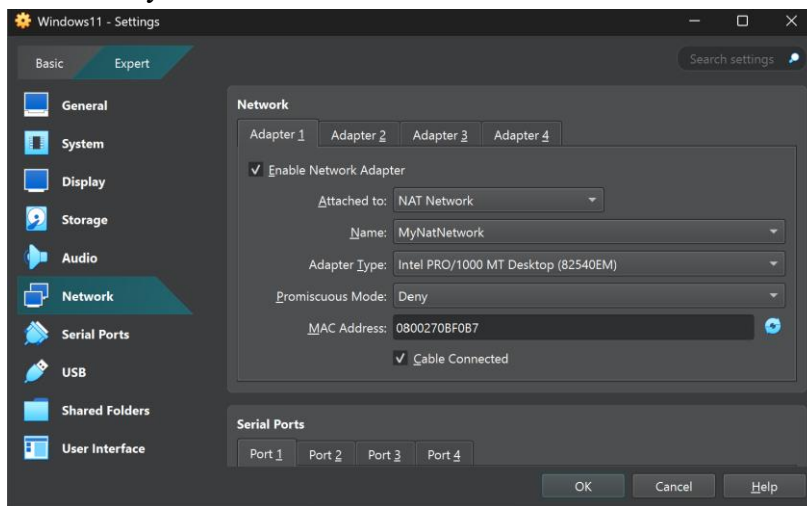
1. Found that IPv4 addresses are same (10.0.2.15) in Windows VM and Kali VM. Shutdown both VMs, within Oracle VirtualBox.
2. Clicked on the Windows 11 VM. In the Oracle Virtualbox, went to the Menu, File >Tools > NetworkManager. Clicked on *NAT Network* tab, and then Clicked on 'Create'.



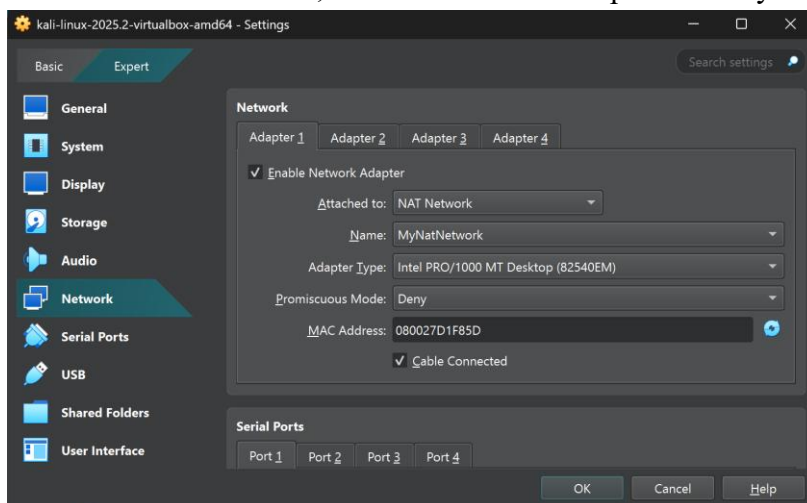
3. Modified the 'NatNetwork' created. Clicked on Properties, then Changed the name to MyNatNetwork. Ensured that 'Enable DHCP' is checked. Then clicked *Apply*.



4. Next, with the Windows VM powered off, clicked on it to highlight it. Clicked on 'Setting'. Clicked on 'Network' under 'Expert' settings. Selected 'NatNetwork'. Selected the Nat Network I created 'MyNatNetwork'. Then Clicked 'OK' to confirm the changes.



5. For the Kali VM, followed the same steps to modify its network settings.



6. Power ON and verified IPv4 addresses in Windows VM and Kali VM are different now.

```
C:\Users\vboxuser>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : v4kuvoelbxsulmmw0zbp1sinuc.gx.internal.cloudapp.net
    Link-local IPv6 Address . . . . . : fe80::ac8f:aff7:66ed:dd50%5
    IPv4 Address. . . . . : 10.0.2.4
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.2.1

(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::c1cf:9e72:107:566 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:d1:f8:5d txqueuelen 1000 (Ethernet)
    RX packets 36 bytes 5794 (5.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 29 bytes 4097 (4.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

STEP 3 – Installing and Testing rsyslog

1. Before installing the *rsyslog*, Update and upgrade the Kali VM.

```
(kali@kali)-[~]
$ sudo apt update
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1189 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali@kali)-[~]
$ sudo apt upgrade
```

2. Found that *rsyslog* is not installed in Kali VM, then installed *rsyslog* using *sudo apt install rsyslog* command.

```
(kali@kali)-[~]
$ sudo apt install rsyslog
[sudo] password for kali:
The following packages were automatically installed and are no longer required:
amass-common libportmidi0 python3-click-plugins
firmware-ti-connectivity libqt5ct-common1.8 python3-gpg
libbluray2 libav1e0.7 python3-kismetcapturebtgeiger
libbson-1.0-0t64 libsfame1 python3-kismetcapturefreaklabszigbee
libgdal36 libsigsegv2 python3-kismetcapturertl433
libgdata-common libsoup-2.4-1 python3-kismetcapturertladsb
libgdata22 libsoup2.4-common python3-kismetcapturertlamr
libgeos3.13.1 libtheora0 python3-packaging-whl
libhdf4-0-alt libtheoradec1 python3-protobuf
libjs-jquery-ui libtheoraenc1 python3-pyinstaller-hooks-contrib
libjs-underscore libudfread0 python3-wheel-whl
libmongoc-1.0-0t64 libvpx9 python3-zombie-imp
libmongocrypt0 libx264-164 samba-ad-dc
libogdi4.1 libyelp0 samba-ad-provision
libplacebo349 python3-bluepy samba-dsdb-modules

Use 'sudo apt autoremove' to remove them.

Installing:
rsyslog
```

3. Verified the installation using *rsyslogd -v* command.

```
(kali@kali)-[~]
$ rsyslogd -v
rsyslogd 8.2506.0 (aka 2025.06) compiled with:
PLATFORM: x86_64-pc-linux-gnu
PLATFORM (lsb_release -d):
FEATURE_REGEX: Yes
GSSAPI Kerberos 5 support: Yes
FEATURE_DEBUG (debug build, slow code): No
32bit Atomic operations supported: Yes
64bit Atomic operations supported: Yes
memory allocator: system default
Runtime Instrumentation (slow code): No
uuid support: Yes
systemd support: Yes
Config file: /etc/rsyslog.conf
PID file: /run/rsyslogd.pid
Number of Bits in RainerScript integers: 64

See https://www.rsyslog.com for more information.
```

4. Used `sudo nano /etc/rsyslog.conf` command to edit the rsyslog configuration file. Modified the configuration file for rsyslog by removing the “#” sign for these lines to provide UDP and TCP syslog reception.

UDP

```
module(load="imudp")
input(type="imudp" port="514")
```

TCP

```
module(load="imtcp")
input(type="imtcp" port="514"
```

Save changes to the configuration file by pressing *Ctrl* + *O* and Enter key, then exit out by pressing *Ctrl* + *X*.



```
(kali㉿kali)-[~]
$ sudo nano /etc/rsyslog.conf
[sudo] password for kali:

kali@kali: ~
Session Actions Edit View Help
GNU nano 8.6 /etc/rsyslog.conf
# /etc/rsyslog.conf configuration file for rsyslog
#
# For more information install rsyslog-doc and see
# /usr/share/doc/rsyslog-doc/html/configuration/index.html
#
#####
### MODULES ###
#####

module(load="imuxsock") # provides support for local system logging
module(load="imklog") # provides kernel logging support
#module(load="immark") # provides --MARK-- message capability

# provides UDP syslog reception
module(load="imudp")
input(type="imudp" port="514")

# provides TCP syslog reception
module(load="imtcp")
input(type="imtcp" port="514")

#####
### GLOBAL DIRECTIVES ###
#####

#
# Set the default permissions for all log files.
#
$FileOwner root
$FileGroup adm
$FileCreateMode 0640
$DirCreateMode 0755
$Umask 0022

#
# Where to place spool and state files
#
$WorkDirectory /var/spool/rsyslog

#
Write to File: /etc/rsyslog.conf
? Help      ?-D DOS Format  ?-A Append      ?-B Backup File  ?-M Mac Format  ?-C Discard buffer  ?- Browse
?C Cancel    ?-M Mac Format  ?-P Prepend    ?-C Discard buffer
```

5. Installed the uncomplicated firewall package, `ufw`, used to manage connections to ports by using `sudo apt install ufw` command.

```

kali@kali:~$ sudo apt install ufw
The following packages were automatically installed and are no longer required:
amass-common libgeo3.13.1 libplacebo349 libsoup2.4-common libyelp0 python3-kismetcaptureertladsb samba-ad-dc
firmware-ti-connectivity libhd4.0-alt libportmidi0 libtheora0 python3-bluepy python3-kismetcaptureertlarm samba-ad-dc
libbluray2 libjs-jquery-ui libtst-common1.8 libtheoradec1 python3-click-plugins python3-kismetcaptureertlarm samba-ad-dc
libb1.0-0t64 libjs-underscore libwavio.7 libtheoracenc1 python3-gpg python3-packaging-whl samba-dsdb-modules
libgd36 libmongoc-1.0-0t64 libfsm1 libudfread0 python3-kismetcaptureertbgeiger python3-pyinstaller-hooks-contrib
libdata-common libmongocrypt0 libsigsev2 libvpx9 python3-kismetcaptureertfreaklabszigbee python3-pyinstaller-hooks-contrib
libdata22 libogdi4.1 libsoup-2.4-1 libx264-164 python3-kismetcaptureertl433 python3-zombie-imp

Use 'sudo apt autoremove' to remove them.

Installing:
ufw

```

6. Started rsyslog using `sudo systemctl start rsyslog` command.

```
(kali㉿kali)-[~]
$ sudo systemctl start rsyslog
```

7. Checked the status of rsyslog to see if it is running using `sudo systemctl status rsyslog` command.

```
(kali㉿kali)-[~]
$ sudo systemctl status rsyslog
● rsyslog.service - System Logging Service
   Loaded: loaded (/usr/lib/systemd/system/rsyslog.service; enabled; preset: enabled)
   Active: active (running) since Sun 2025-10-19 01:27:05 EDT; 2min 31s ago
     Invocation: 8e43369ea3324ce983b1b82e673d06b3
   TriggeredBy: ● syslog.socket
      Docs: man:rsyslogd(8)
            man:rsyslog.conf(5)
            https://www.rsyslog.com/doc/
   Main PID: 2067 (rsyslogd)
      Tasks: 8 (limit: 4456)
     Memory: 1.6M (peak: 2.3M)
        CPU: 113ms
   CGroup: /system.slice/rsyslog.service
            └─2067 /usr/sbin/rsyslogd -n -iNONE

Oct 19 01:27:05 kali systemd[1]: Starting rsyslog.service - System Logging Service ...
Oct 19 01:27:05 kali rsyslogd[2067]: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd. [v8.2506.0]
Oct 19 01:27:05 kali rsyslogd[2067]: [origin software="rsyslogd" swVersion="8.2506.0" x-pid="2067" x-info="https://www.rsyslog.com"] start
Oct 19 01:27:05 kali systemd[1]: Started rsyslog.service - System Logging Service.
```

8. If the rsyslog is not running, we can use `sudo systemctl restart rsyslog` command to restart the rsyslog.

9. Allowed traffic to TCP and UDP port 514 by running `sudo ufw allow 514/tcp` and `sudo ufw allow 514/udp` commands.

```
(kali㉿kali)-[~]
$ sudo ufw allow 514/tcp
Rules updated
Rules updated (v6)

(kali㉿kali)-[~]
$ sudo ufw allow 514/udp
Rules updated
Rules updated (v6)
```

10. Used `logger -n 127.0.0.1 p 514 "This message is from Hashan to test rsyslog"` command to test to see if rsyslog is logging messages.

```
(kali㉿kali)-[~]
$ logger -n 127.0.0.1 p 514 "This message is from Hashan to test rsyslog"
```

11. Verified rsyslog logged test message using `tail -f /var/log/syslog` command.

```
(kali㉿kali)-[~]
$ tail -f /var/log/syslog
2025-10-19T01:34:46.936028-04:00 kali kernel: 05:34:46.913551 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
2025-10-19T01:34:46.958302-04:00 kali kernel: 05:34:46.932309 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
2025-10-19T01:35:01.222582-04:00 kali CRON[2276]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
2025-10-19T01:35:47.917773-04:00 kali kali p 514 This message is from Hashan to test rsyslog
2025-10-19T01:36:10.868296-04:00 kali kernel: 05:34:46.948419 SHCLX11 Shared Clipboard: Converting VBox formats 'BITMAP' to 'INVALID' for X11 (idxFmtX11=0, fmtX11=0, atomT
```

12. Added me as a new user and used Username – hashan and Password – password.

```
(kali㉿kali)-[~]
$ sudo adduser hashan
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for hashan
Enter the new value, or press ENTER for the default
  Full Name []: Hashan Kodippilige
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
```

STEP 4 – Logging SSH Events

1. Allowed SSH through the firewall using *sudo ufw allow ssh* command and enable and checked the status of the ufw using *sudo ufw enable*, and *sudo ufw status* commands.

```
(kali㉿kali)-[~]
$ sudo ufw allow ssh
Rule added
Rule added (v6)

(kali㉿kali)-[~]
$ sudo ufw enable
Firewall is active and enabled on system startup

(kali㉿kali)-[~]
$ sudo ufw status
Status: active

To               Action            From
--
514/tcp          ALLOW            Anywhere
514/udp          ALLOW            Anywhere
22/tcp          ALLOW            Anywhere
514/tcp (v6)     ALLOW            Anywhere (v6)
514/udp (v6)     ALLOW            Anywhere (v6)
22/tcp (v6)     ALLOW            Anywhere (v6)
```

2. Started SSH service using *sudo systemctl start ssh* command.

```
(kali㉿kali)-[~]
$ sudo systemctl start ssh
```

3. Confirmed that SSH is running using `sudo systemctl status ssh` command.

```
(kali@kali)-[~]
$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: active (running) since Sun 2025-10-19 01:54:43 EDT; 3min 17s ago
  Invocation: 8c9ff792a25148bebc2a8912484057af
     Docs: man:sshd(8)
           man:sshd_config(5)
  Process: 2983 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
 Main PID: 2986 (sshd)
    Tasks: 1 (limit: 4456)
  Memory: 2M (peak: 2.6M)
     CPU: 81ms
  CGroup: /system.slice/ssh.service
          └─2986 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 19 01:54:43 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 19 01:54:43 kali sshd[2986]: Server listening on 0.0.0.0 port 22.
Oct 19 01:54:43 kali sshd[2986]: Server listening on :: port 22.
Oct 19 01:54:43 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

4. From Windows VM connected to Kali VM using `ssh hashan@10.0.2.15` command. Displayed feedback on the authenticity of the machine I am trying to connect to and enter 'yes' to continue the connection.

```
C:\Users\vboxuser>ssh hashan@10.0.2.15
The authenticity of host '10.0.2.15 (10.0.2.15)' can't be established.
ED25519 key fingerprint is SHA256:PJeCeeye7iA/uauW2zHJUuqxkt5vAI6410+4/zTOU7w.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? Y
Please type 'yes', 'no' or the fingerprint: yes
```

5. Used the password – *password* to connect to Kali VM as user *hashan*. Successfully connected to the Kali VM from Windows VM.

```
hashan@10.0.2.15's password:
Linux kali 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.

(hashan@kali)-[~]
$ |
```

6. Used `grep hashan /var/log/auth.log` command to see that auth log in the Kali is captured my login attempt using Windows VM.

```
(kali@kali)-[~]
$ grep hashan /var/log/auth.log
2025-10-19T01:39:45.126994-04:00 kali sudo: kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/adduser hashan
2025-10-19T01:39:45.433666-04:00 kali groupadd[2470]: group added to /etc/group: name=hashan, GID=1001
2025-10-19T01:39:45.454330-04:00 kali groupadd[2470]: group added to /etc/gshadow: name=hashan
2025-10-19T01:39:45.456680-04:00 kali groupadd[2470]: new group: name=hashan, GID=1001
2025-10-19T01:39:45.624836-04:00 kali useradd[2475]: new user: name=hashan, UID=1001, GID=1001, home=/home/hashan, shell=/bin/bash, from=/dev/pts/0
2025-10-19T01:39:59.666281-04:00 kali passwd[2484]: pam_unix(passwd:chauthtok): password changed for hashan
2025-10-19T01:40:55.483914-04:00 kali chfn[2488]: changed user 'hashan' information
2025-10-19T01:41:01.296873-04:00 kali gpasswd[2497]: members of group users set by root to kali,hashan
2025-10-19T02:19:35.702168-04:00 kali sshd-session[3214]: Accepted password for hashan from 10.0.2.4 port 49710 ssh2
2025-10-19T02:19:35.708774-04:00 kali sshd-session[3214]: pam_unix(sshd:session): session opened for user hashan(uid=1001) by hashan(uid=0)
2025-10-19T02:19:35.745217-04:00 kali systemd-logind[501]: New session '15' of user 'hashan' with class 'user' and type 'tty'.
2025-10-19T02:19:35.908208-04:00 kali (systemd): pam_unix(systemd-user:session): session opened for user hashan(uid=1001) by hashan(uid=0)
2025-10-19T02:19:35.915560-04:00 kali systemd-logind[501]: New session '16' of user 'hashan' with class 'manager' and type 'unspecified'.
```

STEP 5 – Creating Custom Rules

1. Created a configuration file name *20-hashan-ssh-filter.conf* in Kali VM directory path */etc/rsyslog.d/*. Used *sudo nano /etc/rsyslog.d/20-hashan-ssh-filter.conf* command and this file is for managing certain ssh events.

```
(kali@kali)-[~]  
$ sudo nano /etc/rsyslog.d/20-hashan-ssh-filter.conf  
[sudo] password for kali:
```

2. In the conf file, added the rule to capture failed 'sshd' log in events. Used following command to generate the rule.

```
:msg, contains, "Failed password" /var/log/ssh-failed.log  
&stop
```



```
kali@kali: ~  
Session Actions Edit View Help  
GNU nano 8.6 /etc/rsyslog.d/20-hashan-ssh-filter.conf  
  
:msg, contains, "Failed password" /var/log/ssh-failed.log  
&stop  
  
[ Read 52 lines ]  
[ Location ] [ Undo ] [ Set Mark ] [ To Bracket ] [ Previous ] [ Back ]  
[ Go To Line ] [ Redo ] [ Copy ] [ Where Was ] [ Next ] [ Forward ]
```

3. Restarted rsyslog using *sudo systemctl restart rsyslog* command.

```
(kali@kali)-[~]  
$ sudo systemctl restart rsyslog
```

4. To test the custom configuration used *logger -p auth.warning -t sshd 'Failed password'* command.

```
(kali@kali)-[~]  
$ logger -p auth.warning -t sshd 'Failed password'
```

5. Viewed the log using *cat /var/log/ssh-failed.log* command.

```
(kali@kali)-[~]  
$ cat /var/log/ssh-failed.log  
2025-10-19T03:21:05.698677-04:00 kali sshd: Failed password
```

Step 6 - Accessing Kali Using the Windows VM

1. Checked the SSH status in Kali VM using `sudo systemctl status ssh` command. If it is not running, we can use `sudo systemctl start ssh` command to start ssh service.

```
(kali@kali)-[~]
└─$ sudo systemctl status ssh
[sudo] password for kali:
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: active (running) since Sun 2025-10-19 02:08:27 EDT; 1h 26min ago
 Invocation: b613bf6a654543e2b341d2cdd5317b25
    Docs: man:sshd(8)
          man:sshd_config(5)
   Process: 3123 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 3126 (sshd)
    Tasks: 1 (limit: 4456)
   Memory: 3.8M (peak: 24.1M)
      CPU: 250ms
   CGroup: /system.slice/ssh.service
           └─3126 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 19 02:08:27 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Oct 19 02:08:27 kali sshd[3126]: Server listening on 0.0.0.0 port 22.
Oct 19 02:08:27 kali sshd[3126]: Server listening on :: port 22.
Oct 19 02:08:27 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Oct 19 02:19:35 kali sshd-session[3214]: Accepted password for hashan from 10.0.2.4 port 49710 ssh2
Oct 19 02:19:35 kali sshd-session[3214]: pam_unix(sshd:session): session opened for user hashan(uid=1001) by hashan(uid=0)
```

2. From Windows VM connected to Kali VM using `ssh hashan@10.0.2.15` command. Used wrong passwords for several times and then use the correct password – `password` to connect to Kali VM as user `hashan`.

```
C:\Users\vboxuser>ssh hashan@10.0.2.15
hashan@10.0.2.15's password:
Permission denied, please try again.
hashan@10.0.2.15's password:
Permission denied, please try again.
hashan@10.0.2.15's password:
hashan@10.0.2.15: Permission denied (publickey,password).

C:\Users\vboxuser>ssh hashan@10.0.2.15
hashan@10.0.2.15's password:
Linux kali 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Oct 19 03:40:11 2025 from 10.0.2.4
(kali@kali)-[~]
└─$ |
```

3. Viewed the log using `cat /var/log/ssh-failed.log` command, and found that all the failed logins were captured by `ssh-failed.log` file.

```
(kali@kali)-[~]
└─$ sudo cat /var/log/ssh-failed.log
2025-10-19T03:21:05.698677-04:00 kali sshd: Failed password
2025-10-19T04:11:28.237004-04:00 kali sshd-session[5168]: Failed password for hashan from 10.0.2.4 port 49803 ssh2
2025-10-19T04:11:31.293435-04:00 kali sshd-session[5168]: Failed password for hashan from 10.0.2.4 port 49803 ssh2
2025-10-19T04:14:30.311171-04:00 kali sshd: Failed password
2025-10-19T04:15:28.764982-04:00 kali sshd-session[5245]: Failed password for hashan from 10.0.2.4 port 49804 ssh2
2025-10-19T04:15:32.308227-04:00 kali sshd-session[5245]: Failed password for hashan from 10.0.2.4 port 49804 ssh2
2025-10-19T04:15:37.223039-04:00 kali sshd-session[5245]: Failed password for hashan from 10.0.2.4 port 49804 ssh2
```

4. Used *grep hashan /var/log/auth.log* command to see that auth log in the Kali is captured my failed and successful login attempts using Windows VM.

```
(kali㉿kali)-[~]
$ sudo grep hashan /var/log/auth.log
2025-10-19T01:39:45.126994-04:00 kali sudo:    kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/sbin/adduser hashan
2025-10-19T01:39:45.433666-04:00 kali groupadd[2470]: group added to /etc/group: name=hashan, GID=1001
2025-10-19T01:39:45.454330-04:00 kali groupadd[2470]: group added to /etc/gshadow: name=hashan
2025-10-19T01:39:45.456680-04:00 kali groupadd[2470]: new group: name=hashan, GID=1001
2025-10-19T01:39:45.624836-04:00 kali useradd[2475]: new user: name=hashan, UID=1001, GID=1001, home=/home/hashan, shell=/bin/bash, from=/dev/pts/0
2025-10-19T01:39:59.666281-04:00 kali passwd[2484]: pam_unix(passwd:chauthtok): password changed for hashan
2025-10-19T01:40:55.483914-04:00 kali chfn[2488]: changed user 'hashan' information
2025-10-19T01:41:01.296873-04:00 kali gpasswd[2497]: members of group users set by root to kali,hashan
2025-10-19T02:19:35.702168-04:00 kali sshd-session[3214]: Accepted password for hashan from 10.0.2.4 port 49710 ssh2
2025-10-19T02:19:35.708774-04:00 kali sshd-session[3214]: pam_unix(sshd:session): session opened for user hashan(uid=1001) by hashan(uid=0)
2025-10-19T02:19:35.745217-04:00 kali systemd-logind[501]: New session '15' of user 'hashan' with class 'user' and type 'tty'.
2025-10-19T02:19:35.908208-04:00 kali (systemd): pam_unix(systemd-user:session): session opened for user hashan(uid=1001) by hashan(uid=0)
2025-10-19T02:19:35.915560-04:00 kali systemd-logind[501]: New session '16' of user 'hashan' with class 'manager' and type 'unspecified'.
2025-10-19T02:29:59.880562-04:00 kali sudo:    hashan : user NOT in sudoers ; TTY=pts/0 ; PWD=/home/hashan ; USER=root ; COMMAND=/usr/bin/grep hashan /var/log/auth.log
2025-10-19T02:30:44.747930-04:00 kali sudo:    hashan : user NOT in sudoers ; TTY=pts/0 ; PWD=/home/hashan ; USER=root ; COMMAND=/usr/bin/grep hashan /var/log/auth.log
2025-10-19T02:41:13.133336-04:00 kali sshd-session[3263]: Disconnected from user hashan 10.0.2.4 port 49710
2025-10-19T02:41:13.133492-04:00 kali sshd-session[3214]: pam_unix(sshd:session): session closed for user hashan
2025-10-19T02:41:23.636832-04:00 kali (sd-pam): pam_unix(systemd-user:session): session closed for user hashan
2025-10-19T02:49:12.602894-04:00 kali sudo:    kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/nano /etc/rsyslog.d/20-hashan-ssh-filter.conf
2025-10-19T03:08:26.687507-04:00 kali sudo:    kali : TTY=pts/1 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/nano /etc/rsyslog.d/20-hashan-ssh-filter.conf
2025-10-19T03:40:10.783787-04:00 kali sshd-session[4298]: Accepted password for hashan from 10.0.2.4 port 49732 ssh2
2025-10-19T03:40:10.789300-04:00 kali sshd-session[4298]: pam_unix(sshd:session): session opened for user hashan(uid=1001) by hashan(uid=0)
2025-10-19T03:40:10.810246-04:00 kali systemd-logind[501]: New session '31' of user 'hashan' with class 'user' and type 'tty'.
2025-10-19T03:40:11.021245-04:00 kali (systemd): pam_unix(systemd-user:session): session opened for user hashan(uid=1001) by hashan(uid=0)
2025-10-19T03:40:11.025384-04:00 kali systemd-logind[501]: New session '32' of user 'hashan' with class 'manager' and type 'unspecified'.
2025-10-19T03:40:58.422305-04:00 kali sshd-session[4335]: Disconnected from user hashan 10.0.2.4 port 49732
2025-10-19T03:40:58.450548-04:00 kali sshd-session[4298]: pam_unix(sshd:session): session closed for user hashan
2025-10-19T03:41:26.122352-04:00 kali unix_chkpwd[4401]: password check failed for user (hashan)
2025-10-19T03:41:26.122352-04:00 kali sshd-session[4399]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.4 user=hashan
2025-10-19T03:41:26.123777-04:00 kali sshd-session[4399]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'hashan')
2025-10-19T03:41:28.749698-04:00 kali sshd-session[4399]: Failed password for hashan from 10.0.2.4 port 49736 ssh2
2025-10-19T03:41:33.461803-04:00 kali unix_chkpwd[4402]: password check failed for user (hashan)
2025-10-19T03:41:33.466726-04:00 kali sshd-session[4399]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'hashan')
2025-10-19T03:41:35.377819-04:00 kali sshd-session[4399]: Failed password for hashan from 10.0.2.4 port 49736 ssh2
2025-10-19T03:41:39.287940-04:00 kali unix_chkpwd[4403]: password check failed for user (hashan)
2025-10-19T03:41:39.290184-04:00 kali sshd-session[4399]: pam_winbind(sshd:auth): internal module error (retval = PAM_AUTHINFO_UNAVAIL(9), user = 'hashan')
2025-10-19T03:41:41.094010-04:00 kali sshd-session[4399]: Failed password for hashan from 10.0.2.4 port 49736 ssh2
2025-10-19T03:41:42.091614-04:00 kali sshd-session[4399]: Connection reset by authenticating user hashan 10.0.2.4 port 49736 [preauth]
2025-10-19T03:41:58.625594-04:00 kali sshd-session[4404]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=10.0.2.4 user=hashan
2025-10-19T03:41:58.625142-04:00 kali sshd-session[4404]: Accepted password for hashan from 10.0.2.4 port 49737 ssh2
2025-10-19T03:41:58.696475-04:00 kali unix(sshd:session): session opened for user hashan(uid=1001) by hashan(uid=0)
2025-10-19T03:41:58.696475-04:00 kali systemd-logind[501]: New session '33' of user 'hashan' with class 'user' and type 'tty'.
2025-10-19T03:41:58.857955-04:00 kali (systemd): pam_unix(systemd-user:session): session opened for user hashan(uid=1001) by hashan(uid=0)
```

5. In Kali VM, ran the *last* command and captured the results. The *last* command retrieves information about previous user logging and system reboots by reading data from */var/log/wtmp* file. This file records successful login attempts, logouts, and system reboots with details of username, terminal type (tty or pts), source IP address, login time, logout time, and session duration.

```
(kali㉿kali)-[~]
$ last
hashan pts/0 10.0.2.4 Sun Oct 19 04:17 - still logged in
hashan pts/0 10.0.2.4 Sun Oct 19 04:04 - 04:06 (00:02)
hashan pts/0 10.0.2.4 Sun Oct 19 03:51 - 04:04 (00:12)
hashan pts/0 10.0.2.4 Sun Oct 19 03:41 - 03:51 (00:09)
hashan pts/0 10.0.2.4 Sun Oct 19 03:40 - 03:40 (00:00)
hashan pts/0 10.0.2.4 Sun Oct 19 02:19 - 02:41 (00:21)
kali tty7 :0 Sun Oct 19 01:13 - still logged in
lightdm tty7 :0 Sun Oct 19 01:13 - 01:13 (00:00)
kali tty7 :0 Sat Oct 18 19:51 - 19:51 (00:00)
lightdm tty7 :0 Sat Oct 18 19:50 - 19:51 (00:00)
postgres pts/2 Sat Oct 18 19:33 - 19:33 (00:00)
kali Sat Oct 18 19:24 - still logged in
kali tty7 :0 Sat Oct 18 18:29 - 19:48 (01:19)
lightdm tty7 :0 Sat Oct 18 18:28 - 18:29 (00:00)
kali tty7 :0 Sat Oct 18 16:47 - 17:27 (00:39)
lightdm tty7 :0 Sat Oct 18 16:47 - 16:47 (00:00)
postgres pts/2 Tue Aug 19 16:35 - 16:35 (00:00)
kali Tue Aug 19 16:30 - still logged in
kali tty7 :0 Tue Aug 19 16:09 - still logged in
lightdm tty7 :0 Tue Aug 19 16:09 - 16:09 (00:00)
postgres Thu May 29 15:22 - 15:22 (00:00)

wtmpdb begins Thu May 29 15:22:05 2025
```

When correlating the entries in `/var/log/ssh.failed.log`, `/var/log/auth.log` with `last` command give a lot of information. The `last` command output does not contain the details of failed login attempts, and all the successful login attempts in the `/var/log/auth.log` can be seen in the `last` command output with correct time stamps.

Conclusion

This lab helps to cover the practical aspects of how the syslog is implemented in Kali VM through rsyslog, and how to set up SSH custom rules and capture remote logging events. Configuring firewalls (ufw), creating a user account for logging from Windows VM, and SSH custom filters to set up capture the unsuccessful logging events from this account provide valuable information related to the forensic report.

Creating automated messages of failed logging attempts will provide administrators with a compliant mechanism to protect system integrity and meet audit requirements. Integrating syslog with security controls like ufw and ssh demonstrates the importance of layered defence strategies and intrusion detection workflow. Using these logging events, the analyst or system administrators can get the details of brute force attacks and configuration anomalies. This lab demonstrates that installing syslog not only provides details of logging events but also provides accountability and integrity of the network.

References

1. Andress, J. (2019). *Foundations of Information Security*. No Starch Press
2. Easttom, C. (2023). *Computer Security Fundamentals* (5th ed.). Pearson Education.
3. Sanders, C. (2017). *Practical packet analysis* (3rd ed.).
4. Weiss, S. (2025). *System Programming in Linux*. No Starch Press
5. Chapple, M., Stewart, J. M. *ISC2 CISSP Certified Information Systems Security Professional Official Study Guide* (10th ed.). Sybex